

Evidence-Bound Recovery for Ambiguous Submissions in Local Generative-AI Pipelines

Tool Description

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Abstract—A lost submission response can leave a generation client uncertain whether its backend executed a request. We describe evidence-bound recovery tooling for local image-generation pipelines: a versioned evidence exporter, a same-run submission guard, and controlled fault injection with backend observation. The exporter separates reported state, interpretation and oracle evidence. The guard preserves explicitly unknown outcomes without claiming automatic reconciliation. Four fixed Windows/ComfyUI control and response-loss cases were oracle-evaluable. Each control had one submission and one observed execution; each injected case had an unresolved first call and a resolved second call, with two submissions and executions. The second call created a fresh run with a different seed after observed completion. The demonstration establishes bounded execution visibility, not recovery of the original run or fewer executions. Deployment rates and output-quality improvement remain unmeasured.

I. INTRODUCTION

Consider a client that submits an image-generation workflow and loses the response after the backend accepts it. The backend may continue while the client lacks the returned job identifier. A second submission creates another opportunity for execution, but a timeout or a count of HTTP requests cannot establish what the backend actually did. The relevant question is which observations justify the next recovery action and the subsequent report of that action.

Uncertain outcomes are an established distributed-systems problem. Remote procedure call designs distinguish call semantics under communication failure [1], while idempotent API designs use caller-provided identifiers and explicit service behavior to make retries safe [2]. A local workstation does not remove the distinction between request transmission and backend execution. Our tool addresses how that distinction is represented and inspected in a particular local generation pipeline; it does not introduce a new exactly-once protocol.

`local-gpu-imagegen` combines persistent run records, generation orchestration, a ComfyUI adapter and artifact handling. We describe tooling that preserves the client’s evidence, applies versioned interpretation rules, and records backend observations independently of the client’s returned status. Its product guard is deliberately narrower: an explicitly unknown submission blocks further submission through the same unresolved run at the reviewed entry points.

The tool description has three parts: an evidence representation and read-only exporter; a same-run guard whose unresolved state remains visible; and CPU and Windows fault-observation workflows with separate evidence boundaries.

The Windows demonstration tests visibility across the client-backend boundary. It does not establish that the guard reduces execution counts in that protocol. The contribution is an inspectable integration of evidence interpretation, run-scoped control and backend observation. Section 6 compares its scope with idempotent APIs, workflow execution and tracing. The contribution is scoped to this implementation and evaluated failure windows; no first-of-its-kind claim is made.

II. FAILURE MODEL AND EVIDENCE SEMANTICS

A **product call** is an invocation by the research client. A **run** is the persistent product object created by `local_gpu_start_run`. A **submission** in the Windows results is a proxy-observed `POST /prompt`; a call that fails before that boundary need not produce a submission. An **execution instance** is an observation bound to backend execution events. CPU fixtures instead observe the synthetic worker entry. These units are not interchangeable.

F00 is a normal control. F02 suppresses the first accepted submission response, creating a window in which the client lacks an acknowledged job identifier. F03, covered by the CPU fixtures rather than the Windows campaign reported here, concerns loss of completion information after a job identifier is known. An explicit rejection and a missing response have different evidential meanings. Neither should be silently converted into a backend execution outcome.

The evidence model distinguishes reported state, interpreted state and oracle state. In addition, the Windows research client emits its own call-level labels. Its `resolved` label is derived from a generated product result after checking for artifact hashes. Its `unresolved` label can arise from several client errors. Thus a research-client label alone does not prove that a durable product manifest has a particular recovery state. In F02, the proxy’s accepted-response loss and the oracle binding provide the additional context.

III. TOOL DESIGN

A. Evidence-preserving export

The exporter accepts a record or run manifest and produces a separate normalized record. It copies reported state, records the mapping version and reasons, and retains supplied oracle state separately. The schema includes source identity, request and input digests, backend identity, job identifier, artifact

hash and validator version. Missing fields carry explicit reasons instead of fabricated values. The inspected mapping is `research-normalization-v2`.

Interpretation checks more than an execution-success label. Job and artifact bindings, approval and validation evidence, and recovery state constrain the `execution_verified` decision. An unresolved recovery state prevents this flag from becoming true even when other evidence reports successful execution. This permits the record to express backend completion without asserting that the product’s recovery obligation has been discharged. The exporter does not invoke the engine, retry a request or update the source run. Its output writer uses exclusive file creation. These are implementation properties, not a claim of complete filesystem security.

B. Using the evidence record

An investigator first inspects the run manifest and the recorded submission outcome. If submission is explicitly unknown, inspection must preserve that uncertainty: the missing job identifier must not be synthesized from a later successful call. Backend observation is then associated with the relevant request and execution instance, while artifact checks retain their own result. The export command writes a new record and refuses to overwrite an existing export:

```
python -m
scripts.research.export_records
-input run-record.json -output
evidence.json
```

The exported record lets a reader inspect reported state, interpreted state, mapping reasons and oracle evidence without invoking recovery. It does not acquire a missing oracle automatically. The Windows observer export and the normalized exporter schema are separate interfaces; a complete automatic conversion between them is not established by the pilot.

For example, an unresolved run and successful backend evidence may coexist. The execution field can describe the observed backend outcome while recovery remains required and execution verification remains false. This separation is the intended diagnostic output, rather than an error to hide by replacing the original run state with the latest call’s result.

C. Recovery guard and its scope

`AssetRunEngine` and `RunStore` preserve the explicit transport marker `submission_outcome=unknown` as an unresolved attempt. The reviewed submission entry checks that unresolved attempt before permitting further generation. The guard applies within the same run, including attempts with a different idempotency key. The corresponding inspection action is `get_run`.

This guard does not discover a missing backend job identifier or reconcile an unknown job automatically. A newly created run lies outside the prior run’s guard. Existing recovery for a known job in a two-stage operation is a separate path and is not attributed to this no-job-ID guard. The resulting tradeoff is explicit: the guard withholds another submission through that run while the required evidence is unavailable; it does not guarantee eventual completion.

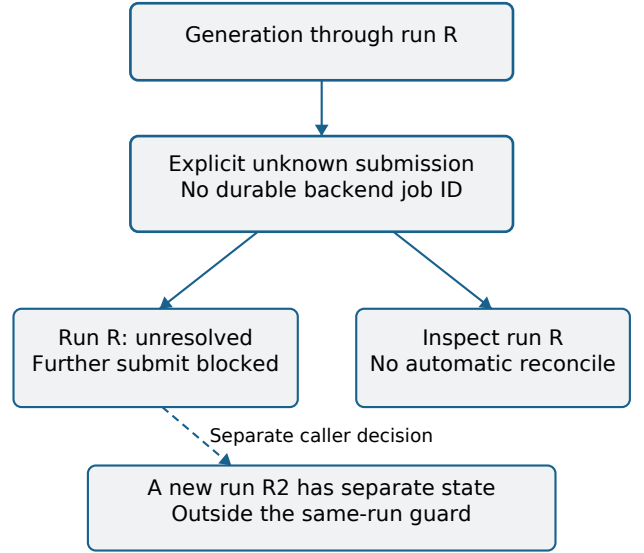


Fig. 1. Scope of the product guard, derived from the reviewed store and engine entry points. A new run has separate state. Dashed arrows denote a separate caller decision, not automatic reconciliation.

D. Controlled observation on Windows

Figure 1 shows the product and research boundaries. The research harness adds a transport shim and loopback fault proxy. The shim routes prompt transport through the proxy while retaining the actual backend endpoint in the product’s identity client. F02 drops the first response only after backend acceptance. This arrangement is a research instrument, not a native ComfyUI recovery feature.

The observer connects before the product invocation and uses the operation key as its ComfyUI client identifier, matching the product’s event-routing identity. `ComfyUIEventOracle` requires an execution-start event, a terminal event and completed history bound to a prompt identifier. It derives an execution-instance identifier using the backend boot identity, prompt identifier and observed start sequence. Its independence is from the product return value: it still trusts the backend’s event and history interfaces, and it is not a physical GPU monitor or a crash-durable audit log.

The controller permits the second F02 product call only after the first execution has a bindable completion. The client creates a new run on each call and uses `seed = 4100 + call_index`. This is observation-gated fresh-run generation, not automatic reconciliation of the original run. Matching frozen request digests does not establish full prompt equality: the inspected digest function excludes the seed and does not use its plan argument.

IV. EVIDENCE AND DEMONSTRATION

A. CPU components and deterministic fixtures

The CPU harness separates submission events from synthetic worker-entry events. Its fixtures cover normal operation and response loss before and after job-ID acquisition,

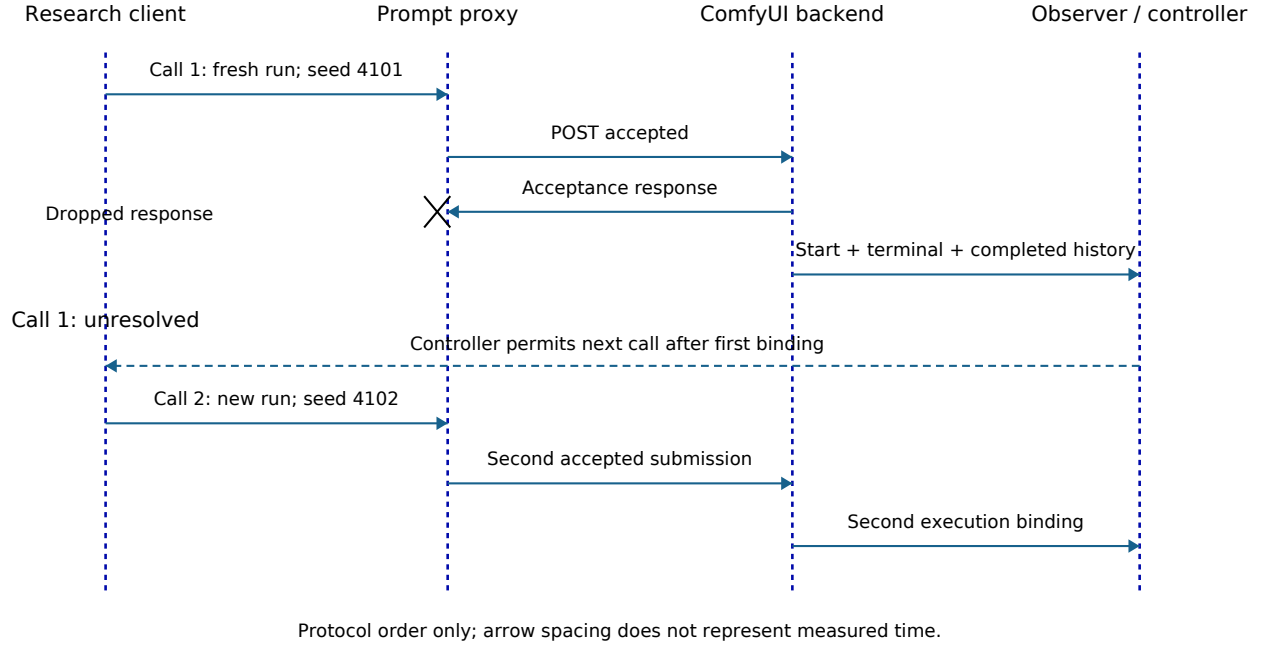


Fig. 2. Controlled F02 protocol. The proxy hides the accepted response; backend observation gates a second call that creates a fresh run with a changed seed. Retained receipts and bindings support the two submissions and the ordering of the second receipt after the first completion binding. Spacing is schematic, not a measured latency or GPU-runtime plot.

including single-stage and two-stage paths. Oracle self-checks exercise cases in which submission and execution counts differ and in which a shared prompt identifier does not imply a single execution. These are fixed semantic checks, not randomly sampled deployments.

Historical command output records 40 research component tests passing in 7.110 seconds using Python unittest discovery, with OK and exit code 0. The preparation report identifies the runtime as Windows Python 3.15.0a8. A later macOS check targeted the transport shim and campaign controller: seven tests passed in 0.003 seconds, followed by successful compilation and whitespace checks. These are distinct stages; the earlier 40-test result is not a full-suite result for the final shim. Neither count is an experimental sample size. Exact immutable source binding of the historical test worktrees remains incomplete.

Table I reports the retained paired CPU matrix. Reanalysis checked all twelve synthetic traces: each execution-start identifier was unique within its case and matched a finish identifier, and request-received counts matched the stored submission totals. The control cases are separate from injected cases. The invalid earlier matrix version, whose single-stage control failed because of a harness result-field mismatch, is excluded.

The distinguishing F02 single-stage fixture has a concrete cost: W3 limits the synthetic executions to one but does not complete the product operation in the observation window. This is evidence of conservative blocking, not a net reliability improvement. The unchanged F03 two-stage row exercises existing known-job recovery. These CPU paths must not be

pooled with the Windows fresh-run protocol in Fig. 2.

B. Four controlled Windows cases

We denote the frozen baseline product revision by B2 and the revision with the same-run ambiguity guard by W3. The endpoint-preserving campaign report records both revisions under F00 and F02. All four fixed cases were oracle-evaluable; the two F00 cases were controls and the two F02 cases had confirmed response-loss injection. Table II reports the observations without converting these counts into a success probability.

The campaign totals were six calls, six proxy submissions and six bound execution instances. The F02 second call used a fresh run and changed seed. The four report records matched the corresponding retained JSONL cases. Their six execution bindings had distinct identifiers, completed-history flags and terminal-event labels. For both F02 cases, the second proxy receipt followed the first completion binding. Full raw-event reconstruction remains unavailable. No rate is estimated.

The report identifies an RTX 5070 Ti Laptop GPU and ComfyUI v0.30.0 at a pinned source revision, with model and workflow hashes checked against the campaign configuration. These are recorded environment facts for this campaign. They do not establish behavior across other models, backend versions or machines.

Both systems produced two observed executions in F02. This demonstrates that the research observer could retain execution evidence despite the first call’s ambiguous outcome. It does not demonstrate an execution-count advantage for W3. Because the second call changed run and seed, the table

TABLE I
DETERMINISTIC CPU FAULT COVERAGE FROM THE RETAINED PAIRED-V2 MATRIX. S/E MEANS SUBMISSIONS/EXECUTIONS. “COMPLETE” IS THE FIXTURE’S VALID-COMPLETION FLAG. THESE ARE TWELVE SYNTHETIC TRACES, NOT DEPLOYMENT SAMPLES.

Fault	Path	B2 S/E	W3 S/E	Complete (B2/W3)	B2 recovery	W3 recovery
F00	Single stage	1/1	1/1	yes/yes	Not needed	Not needed
F00	Two stage	1/1	1/1	yes/yes	Not needed	Not needed
F02	Single stage	2/2	1/1	yes/no	New submission	Unknown; blocked
F02	Two stage	1/1	1/1	no/no	Unknown	Unknown; blocked
F03	Single stage	2/2	2/2	yes/yes	New submission	New submission
F03	Two stage	1/1	1/1	yes/yes	Same job reconciled	Same job reconciled

TABLE II
FOUR FIXED CASES, CHECKED AGAINST RETAINED REPORT FIELDS. S/E ARE PROXY SUBMISSIONS/BOUND EXECUTION INSTANCES; U/R ARE UNRESOLVED/RESOLVED RESEARCH-CLIENT CALLS. ALL FOUR CASES WERE ORACLE-EVALUABLE.

System	Case	Calls	S/E	Call sequence
B2	F00	1	1/1	R
W3	F00	1	1/1	R
B2	F02	2	2/2	U → R
W3	F02	2	2/2	U → R

also does not establish a duplicate execution of an identical generation request.

C. Resolution labels require an explicit aggregation rule

A resolved second call does not erase the first call’s unresolved result or show that its original run was reconciled. Read-only inspection of the retained campaign report confirmed that both F02 records carry `resolved=1` and `unresolved=1`. These are independent any-call flags: across the four cases, `resolved` sums to four and `unresolved` to two. They are overlapping counts, not mutually exclusive outcomes. The earlier Markdown summary’s zero unresolved cases is therefore not the controller aggregation. Table II preserves the call sequence; the evidence addendum records this correction without rewriting the historical report. No recovery-success rate is inferred.

There is also a counting distinction within the observer export. Successive oracle decisions retain cumulative raw-event snapshots, so summing their event counts can count earlier events again. The controller’s execution count uses distinct bound execution-instance identifiers. A raw-evidence audit must check those bindings, not substitute aggregate event counts.

V. REPRODUCIBILITY AND LIMITATIONS

The paired CPU records preserve product revisions, harness digests and event traces. The Windows source and report identities are recorded separately in the companion evidence ledger. The audit and figure-generation scripts consume existing records only; they do not run the generation backend. Reanalysis is reproducible from the included derived records, while recreating the Windows campaign also requires the pinned backend, model, workflow and private setup. A fresh clean-checkout execution of that campaign was not performed.

The retained Windows JSONL records contain sanitized case summaries, execution bindings and aggregate event

counts, rather than full raw WebSocket messages and history snapshots. Thus the audit verifies internal consistency of retained bindings; it cannot independently reconstruct every backend event. The derived, path-redacted evidence extract and a read-only audit script are included with the manuscript source. No backend call is needed to rerun that consistency check.

Read-only hashing found that the four generated-call artifact hashes matched current output image bytes on the Windows host. Each case had one product-output PNG; the unresolved first F02 call recorded no product artifact hash. Therefore these checks do not verify the first F02 execution’s output bytes. The oracle’s path fingerprints hash metadata strings and cannot substitute for content hashes. Neither a matching content hash nor completed history establishes visual quality or final user acceptance.

Observation depends on receiving and binding backend events within the harness window. Missing events limit evaluability; they do not establish non-execution. The same-process CPU observer and backend-origin Windows observer have different failure boundaries, neither of which establishes crash-complete auditability. The fixed case set contains no basis for deployment failure frequency, natural duplicate-execution frequency, performance improvement or broad model robustness. No GPU run, model download or new experiment was performed for this manuscript revision.

VI. RELATED WORK AND DISCUSSION

RPC failure semantics provide the conceptual starting point: a failed communication does not necessarily settle whether a remote operation ran [1]. Idempotent API design addresses safe retries through an explicit contract, including caller-provided request identifiers [2]. Our tool relies on neither a prompt identifier nor a client idempotency key as sufficient evidence of backend deduplication. Its role is to preserve and qualify observations when such a guarantee has not been established for the evaluated path.

The practical distinction is between observation and control. Backend events can establish that work completed while the client’s original run remains unresolved. Conversely, a conservative same-run guard can block a further submission without resolving that uncertainty. The Windows harness crosses a different boundary by creating a new run after observation. Keeping these scopes separate explains why the demonstration can be oracle-evaluable without showing automatic recovery or a W3 reduction in executions.

Reliable State Machines provide a stronger control contract within their programming model: persistent state and runtime-managed atomic processing support failure transparency and exactly-once input processing [3]. Our tool does not place the external generator inside such a transactional runtime. It exposes uncertainty at that boundary and restricts one product entry path, rather than claiming an equivalent processing guarantee.

Temporal’s Activity documentation describes the same lost-completion boundary: a worker may finish an Activity and crash before reporting completion, leading to a retry. It recommends idempotent Activities [4]. This is an important comparison because durable workflow state alone does not eliminate uncertainty about external side effects. Our same-run guard withholds a submission under explicit uncertainty, but supplies no equivalent general workflow-execution or automatic reconciliation guarantee.

OpenTelemetry represents operations through spans and uses trace context to relate observations across a request [5]. Such correlation provides a useful representation for a distributed execution path. In this tool, the additional application-specific rule is which observations permit an execution claim: backend start and terminal events must bind to completed history, and unresolved product recovery can still veto `execution_verified`. Correlation identifiers alone are not used as execution counts.

Toxiproxy provides configurable TCP connection fault injection for testing [6]. Our research proxy uses an application-level acceptance condition to select the response to suppress, while the observer separately checks execution. This comparison does not show that Toxiproxy could not support an equivalent setup; no comparative implementation or usability measurement was performed.

Filibuster’s project overview describes synthesizing fault-injected test variations from developer integration tests to check application error handling [7]. Our fixed case set does not provide that automatic search coverage. It concentrates on evidence interpretation at a selected submission barrier and preserves a separate backend observation channel. The comparison uses the project overview, not a reproduction of Filibuster’s research results.

This is a documentation-based comparison of responsibilities, not a performance benchmark or a feature-absence survey. The practical contribution is making the relationship between client ambiguity, backend observation and permitted claims inspectable for a local generation workflow. The comparisons delimit the tool responsibilities; they do not establish superiority, exhaustive novelty, or measured developer productivity.

VII. CONCLUSION

The tool preserves ambiguous client outcomes alongside separately bound execution evidence and implements a conservative guard within an unresolved run. The controlled Windows cases demonstrate execution visibility under lost acceptance responses, while exposing the distinction between a new successful call and reconciliation of the original run.

These bounded observations support an evidence-preserving tool description. They do not establish automatic unknown-job recovery, deployment-wide reliability or natural duplicate-execution rates.

DATA, CODE AND RESPONSIBLE USE

The manuscript source includes derived Windows evidence, retained CPU-record pointers, a deterministic audit script and figure sources. Models, private configuration, generated images and raw host paths are not redistributed in this package. The analysis concerns controlled software executions; it makes no claim about human participants or image-quality acceptance. Human-subjects or institutional-ethics determinations are not inferred from this scope.

OpenAI Codex assisted with manuscript drafting, source inspection, record consistency checks and figure-generation code. The AI-assisted review is an internal, single-context check and not independent peer review. Human authors must approve the final claims, attribution, declarations and submission.

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